Quadratics & Polynomials

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Factoring quadratics

Solve $x^2 + 7x + 12 = 0$ by factoring.

$(x + 3)(x + 4) = 0$, so $x = -3$ or $x = -4$.

2 [Foundational] • Target: Factoring quadratics

Solve $x^2 - 9x + 20 = 0$ by factoring.

$(x - 4)(x - 5) = 0$, so $x = 4$ or $x = 5$.

3 [Foundational] • Target: Factoring quadratics

Solve $2x^2 + 5x - 3 = 0$ by factoring.

$(2x - 1)(x + 3) = 0$, so $x = \frac{1}{2}$ or $x = -3$.

4 [Foundational] • Target: Vertex form and completing the square

Write $y = x^2 - 6x + 5$ in vertex form by completing the square, and state the vertex.

$x^2 - 6x + 5 = (x - 3)^2 - 9 + 5 = (x - 3)^2 - 4$. Vertex: $(3, -4)$.

5 [Foundational] • Target: Vertex form and completing the square

Solve $x^2 + 8x + 3 = 0$ by completing the square.

$x^2 + 8x = -3$, so $(x + 4)^2 - 16 = -3$, giving $(x + 4)^2 = 13$, so $x = -4 \pm \sqrt{13}$.

6 [Foundational] • Target: The quadratic formula and discriminant

Solve $3x^2 - 5x - 2 = 0$ using the quadratic formula.

$a = 3, b = -5, c = -2$. Discriminant $= 25 + 24 = 49$. $x = \frac{5 \pm 7}{6}$, so $x = 2$ or $x = -\frac{1}{3}$.

7 [Foundational] • Target: The quadratic formula and discriminant

Find the discriminant of $2x^2 + 3x + 5 = 0$ and state the nature of its roots.

Discriminant $= 3^2 - 4(2)(5) = 9 - 40 = -31$, which is negative, so the equation has no real roots.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: The quadratic formula and discriminant

For what values of k does $x^2 + kx + 16 = 0$ have exactly one repeated real root?

A repeated root occurs when the discriminant is zero: $k^2 - 4(1)(16) = 0$, so $k^2 = 64$, giving $k = \pm 8$.

9 [Intermediate] • Target: Graphing parabolas

The graph of $y = x^2 - 2x - 3$ is shown below. Use the graph to state the x-intercepts, then confirm your answer algebraically by factoring.

From the graph, the parabola crosses the x-axis at $x = -1$ and $x = 3$. Algebraically: $x^2 - 2x - 3 = (x - 3)(x + 1) = 0$, so $x = 3$ or $x = -1$.

10 [Intermediate] • Target: Graphing parabolas

The graph of $y = -2x^2 + 8x - 3$ opens which direction, and what is the x-coordinate of its vertex?

Since $a = -2 < 0$, the parabola opens downward. Vertex: $x = -\frac{b}{2a} = -\frac{8}{2(-2)} = 2$.

11 [Intermediate] • Target: Polynomial multiplication and special products

Expand $(2x - 5)^2$.

$(2x - 5)^2 = 4x^2 - 20x + 25$.

12 [Intermediate] • Target: Polynomial multiplication and special products

Multiply $(x + 7)(x - 7)$.

Using the difference-of-squares identity: $x^2 - 49$.

13 [Intermediate] • Target: Polynomial multiplication and special products

Multiply $(3x + 4)(3x - 4)$.

$9x^2 - 16$.

14 [Intermediate] • Target: Polynomial division and the remainder theorem

Use synthetic division to divide $x^3 - 2x^2 - 5x + 6$ by $(x - 1)$, and state the quotient and remainder.

Using synthetic division with root 1 on coefficients $1, -2, -5, 6$: bring down 1; $1(1) = 1$, $-2 + 1 = -1$; $-1(1) = -1$, $-5 + (-1) = -6$; $-6(1) = -6$, $6 + (-6) = 0$. Quotient: $x^2 - x - 6$, remainder: 0.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Polynomial division and the remainder theorem • Cross-Lesson: 3.01 Functions & Graphs

Use the Remainder Theorem to find the remainder when $x^3 + 4x^2 - x + 2$ is divided by $(x + 2)$.

By the Remainder Theorem, the remainder equals $f(-2)$:
$(-2)^3 + 4(-2)^2 - (-2) + 2 = -8 + 16 + 2 + 2 = 12$.

16 [Challenging] • Target: Quadratic word problems • Cross-Lesson: 4.01 Ratios, Rates & Data

A ball is thrown upward so that its height is modeled by $h(t) = -16t^2 + 64t + 5$ (in feet), where t is time in seconds. Find the time at which the ball reaches its maximum height, and find that maximum height.

Maximum occurs at $t = -\frac{b}{2a} = -\frac{64}{2(-16)} = 2$ seconds.
$h(2) = -16(4) + 64(2) + 5 = -64 + 128 + 5 = 69$ feet.

17 [Challenging] • Target: Quadratic word problems • Cross-Lesson: 5.01 Statistics & Probability

The area of a rectangle is $x^2 + 2x - 15$ square meters, and its width is $x - 3$. Find its length in terms of x by factoring.

$x^2 + 2x - 15 = (x + 5)(x - 3)$. Since width $= x - 3$, the length is $x + 5$.

18 [Challenging] • Target: Sum and product of roots • Cross-Lesson: 6.01 Geometry

For $x^2 - 7x + 10 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.

Sum $= -\frac{-7}{1} = 7$. Product $= \frac{10}{1} = 10$.

19 [Challenging] • Target: Sum and product of roots • Cross-Lesson: 7.01 Trigonometry & Additional Topics

A quadratic equation has roots 3 and -5. Write the equation in standard form.

$(x - 3)(x + 5) = x^2 + 2x - 15 = 0$.

20 [Challenging] • Target: A well-designed word problem: maximizing weekly profit • Cross-Lesson: 8.01 Exponentials & Logarithms

This question has three parts. A company's weekly profit (in thousands of dollars) from producing x hundred units is modeled by $P(x) = -2x^2 + 20x - 32$. (a) Factor $P(x)$ to find the break-even production levels, where profit is zero. (b) Find the vertex of the parabola to determine the production level that maximizes profit, and state the maximum profit. (c) State the range of production levels (in hundreds of units) for which profit is positive.

(a) $P(x) = -2(x^2 - 10x + 16) = -2(x - 2)(x - 8)$. Break-even at $x = 2$ and $x = 8$ (hundreds of units). (b) Vertex: $x = -\frac{b}{2a} = -\frac{20}{2(-2)} = 5$. $P(5) = -2(25) + 20(5) - 32 = -50 + 100 - 32 = 18$, so the maximum profit is 18 thousand dollars at $x = 5$ hundred units (500 units). (c) Profit is positive between the roots: $2 < x < 8$, i.e. between 200 and 800 units.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Factoring quadratics

Solve $x^2 - 15x + 54 = 0$ by factoring.

a. $6, 8$

b. $7, 9$

c. $6, 9$ ✓

d. $-6, -9$

$(x - 9)(x - 6) = 0$, so $x = 9$ or $x = 6$.

22 [Foundational] • Target: Vertex form and completing the square

Solve $x^2 + 5x + 4 = 0$ by factoring.

a. $4, 1$

b. $-4, -2$

c. $-3, -1$

d. $-4, -1$ ✓

$(x - (-1))(x - (-4)) = 0$, so $x = -1$ or $x = -4$.

23 [Foundational] • Target: The quadratic formula and discriminant

Solve $x^2 + 4x = 0$ by factoring.

a. $4, 0$

b. $-3, 0$

c. $-4, 0$ ✓

d. $-4, -1$

$(x - 0)(x - (-4)) = 0$, so $x = 0$ or $x = -4$.

24 [Foundational] • Target: Graphing parabolas

Solve $x^2 + 3x + 2 = 0$ by factoring.

a. $2, 1$

b. $-1, -1$

c. $-2, -1$ ✓

d. $-2, -2$

$(x - (-2))(x - (-1)) = 0$, so $x = -2$ or $x = -1$.

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25 [Foundational] • Target: Polynomial multiplication and special products

Solve $x^2 - x - 12 = 0$ by factoring.

a. $-3, 4$ ✓

b. $-2, 4$

c. $-3, 3$

d. $3, -4$

$(x - (-3))(x - 4) = 0$, so $x = -3$ or $x = 4$.

26 [Foundational] • Target: Polynomial division and the remainder theorem

Solve $x^2 + 11x + 28 = 0$ by factoring.

a. $-7, -4$ ✓

b. $-6, -4$

c. $-7, -5$

d. $7, 4$

$(x - (-4))(x - (-7)) = 0$, so $x = -4$ or $x = -7$.

27 [Foundational] • Target: Quadratic word problems

Solve $x^2 - 9x = 0$ by factoring.

a. $0, 8$

b. $0, -9$

c. $0, 9$ ✓

d. $1, 9$

$(x - 0)(x - 9) = 0$, so $x = 0$ or $x = 9$.

28 [Foundational] • Target: Sum and product of roots

Solve $2x^2 - 17x + 30 = 0$ using the quadratic formula.

a. $\frac{5}{2}, 5$

b. $\frac{7}{2}, 6$

c. $\frac{5}{2}, 6$ ✓

d. $-\frac{5}{2}, -6$

$a = 2, b = -17, c = 30$. Discriminant $= (-17)^2 - 4(2)(30) = 289 - 240 = 49$. $x = \frac{17 \pm 7}{4}$, so $x = 6$ or $x = \frac{5}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: maximizing weekly profit

Solve $x^2 - 8x + 12 = 0$ using the quadratic formula.

a. $-2, -6$

b. $3, 6$

c. $2, 5$

d. $2, 6$ ✓

$a = 1, b = -8, c = 12$. Discriminant $= (-8)^2 - 4(1)(12) = 64 - 48 = 16$. $x = \frac{8 \pm 4}{2}$, so $x = 6$ or $x = 2$.

30 [Intermediate] • Target: Factoring & Quadratic Formula

Solve $2x^2 - 6x - 8 = 0$ using the quadratic formula.

a. $-1, 4$ ✓

b. $-1, 3$

c. $0, 4$

d. $1, -4$

$a = 2, b = -6, c = -8$. Discriminant $= (-6)^2 - 4(2)(-8) = 36 + 64 = 100$. $x = \frac{6 \pm 10}{4}$, so $x = 4$ or $x = -1$.

31 [Intermediate] • Target: Vertex Form & Parabola Extremes

Solve $x^2 - 3x - 10 = 0$ using the quadratic formula.

a. $-2, 5$ ✓

b. $-2, 4$

c. $-1, 5$

d. $2, -5$

$a = 1, b = -3, c = -10$. Discriminant $= (-3)^2 - 4(1)(-10) = 9 + 40 = 49$. $x = \frac{3 \pm 7}{2}$, so $x = 5$ or $x = -2$.

32 [Intermediate] • Target: Polynomial Division & Factors

Solve $2x^2 - 6x = 0$ using the quadratic formula.

a. $0, 2$

b. $0, 3$ ✓

c. $0, -3$

d. $1, 3$

$a = 2, b = -6, c = 0$. Discriminant $= (-6)^2 - 4(2)(0) = 36 = 36$. $x = \frac{6 \pm 6}{4}$, so $x = 3$ or $x = 0$.

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33 [Intermediate] • Target: Advanced Polynomial Graphs & Vieta

Solve $3x^2 + 12x + 9 = 0$ using the quadratic formula.

a. $-3, -1$ ✓

b. $-2, -1$

c. $-3, -2$

d. $3, 1$

$a = 3, b = 12, c = 9$. Discriminant $= (12)^2 - 4(3)(9) = 144 - 108 = 36$. $x = \frac{-12 \pm 6}{6}$, so $x = -1$ or $x = -3$.

34 [Intermediate] • Target: Factoring quadratics

Solve $2x^2 - 14x + 12 = 0$ using the quadratic formula.

a. $-1, -6$

b. $2, 6$

c. $1, 6$ ✓

d. $1, 5$

$a = 2, b = -14, c = 12$. Discriminant $= (-14)^2 - 4(2)(12) = 196 - 96 = 100$. $x = \frac{14 \pm 10}{4}$, so $x = 6$ or $x = 1$.

35 [Intermediate] • Target: Vertex form and completing the square

Find the vertex of the parabola $y = -x^2 + 16x - 71$.

a. $(10, -7)$

b. $(8, -7)$ ✓

c. $(8, 7)$

d. $(-8, -7)$

$h = -\frac{b}{2a} = -\frac{16}{-2} = 8$. $k = y(8) = -1(8)^2 + 128 - 71 = -7$. Vertex: $(8, -7)$.

36 [Intermediate] • Target: The quadratic formula and discriminant

Find the vertex of the parabola $y = -3x^2 - 24x - 47$.

a. $(-2, 1)$

b. $(-4, 1)$ ✓

c. $(-4, -1)$

d. $(4, 1)$

$h = -\frac{b}{2a} = -\frac{-24}{-6} = -4$. $k = y(-4) = -3(-4)^2 + 96 - 47 = 1$. Vertex: $(-4, 1)$.

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37 [Intermediate] • Target: Graphing parabolas

Multiply $(2x + 11)^2$.

a. $4x^2 - 44x + 121$

b. $4x^2 + 44x - 121$

c. $4x^2 + 44x + 121$ ✓

d. $4x^2 + 121$

$(2x + 11)^2 = 4x^2 + 44x + 121$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Polynomial multiplication and special products • Cross-Lesson: 6.01 Geometry

Multiply $(4x + 9)^2$.

a. $16x^2 + 81$

b. $16x^2 + 72x + 81$ ✓

c. $16x^2 + 72x - 81$

d. $16x^2 - 72x + 81$

$(4x + 9)^2 = 16x^2 + 72x + 81$.

39 [Challenging] • Target: Polynomial division and the remainder theorem • Cross-Lesson: 7.01 Trigonometry & Additional Topics

For $2x^2 - 13x + 8 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.

a. $\frac{13}{2}, -4$

b. $-\frac{13}{2}, 4$

c. $\frac{13}{2}, 4$ ✓

d. $4, \frac{13}{2}$

Sum $= -\frac{-13}{2} = \frac{13}{2}$. Product $= \frac{8}{2} = 4$.

40 [Challenging] • Target: Quadratic word problems • Cross-Lesson: 8.01 Exponentials & Logarithms

For $x^2 + 6x + 6 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.

a. $6, -6$

b. $6, 6$

c. $-6, 6$ ✓

d. $-6, -6$

Sum $= -\frac{6}{1} = -6$. Product $= \frac{6}{1} = 6$.

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41 [Challenging] • Target: Factoring & Quadratic Formula • Cross-Lesson: 1.01 Heart of Algebra

What is the positive solution to the quadratic equation $3x^2 - 10x - 8 = 0$?

a. 2

b. 4 ✓

c. $\frac{2}{3}$

d. $\frac{4}{3}$

Factor by grouping: $(3x + 2)(x - 4) = 0$. The roots are $x = -\frac{2}{3}$ and $x = 4$. The positive solution is 4.

42 [Challenging] • Target: Factoring & Quadratic Formula • Cross-Lesson: 3.01 Functions & Graphs

What is the sum of the solutions of the quadratic equation $2x^2 + 8x - 15 = 0$?

a. -4 ✓

b. 4

c. -7.5

d. 8

By Vieta's formula, sum of roots is $-\frac{b}{a} = -\frac{8}{2} = -4$.

43 [Challenging] • Target: Vertex Form & Parabola Extremes • Cross-Lesson: 4.01 Ratios, Rates & Data

What is the axis of symmetry of the parabola $y = x^2 + 8x - 5$?

a. $x = -4$ ✓

b. $x = 4$

c. $x = -8$

d. $y = -4$

Axis of symmetry is $x = -\frac{b}{2a} = -\frac{8}{2(1)} = -4$.

44 [Challenging] • Target: Polynomial Division & Factors • Cross-Lesson: 5.01 Statistics & Probability

If $(x - 3)$ is a factor of $x^3 - 2x^2 - kx + 12$, what is the value of k?

a. 3

b. 4

c. 7 ✓

d. 9

Factor means $P(3) = 0$:

$3^3 - 2(3^2) - 3k + 12 = 0 \Rightarrow 27 - 18 - 3k + 12 = 0 \Rightarrow 21 - 3k = 0 \Rightarrow 3k = 21 \Rightarrow k = 7$.

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45 [Challenging] • Target: Advanced Polynomial Graphs & Vieta • Cross-Lesson: 6.01 Geometry

What are all real roots of the polynomial equation $x^4 - 5x^2 + 4 = 0$?

a. $x = \pm 1, \pm 2$ ✓

b. $x = \pm 1, \pm 4$

c. $x = 1, 2$

d. $x = 0, 1, 4$

Factor as quadratic in x^2: $(x^2 - 1)(x^2 - 4) = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$, and $x^2 = 4 \Rightarrow x = \pm 2$.

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